Optimal Canonical factorization of multiblock data

Jean-Michel Galharret[1] & Stephanie Bougeard [2] & Katrijn Van Deun[3] & Mohamed Hanafi[1]

[1] *Oniris, INRAE, StatSC, Nantes, France.*
[2] *Epidemiology and Welfare, ANSES, Ploufragan, France.*
[3] *Department of Methodology and Statistics, Tilburg University, Tilburg, The Netherlands.*

1 Introduction

Data sets structured into blocks of variables measured on the same set of observations are frequently encountered in practice. These blocks may consist of quantitative variables originating from distinct measurement sources or correspond to thematic groupings defined by the user. For instance, in omics studies, blocks typically comprise variables derived from different analytical techniques (e.g., proteomics, metabolomics). In psychology, multiblock data structures commonly arise when multiple sets of variables are collected on the same individuals to capture different psychological domains. For example, distinct blocks may include self-report questionnaire scores assessing personality traits, behavioral task performance measures, and neurophysiological indicators such as electroencephalography features.

The four expected objectives of exploratory multiblock data analysis are: (i) to perform a joint dimensionality reduction, (ii) to examine relationships among blocks, (iii) to investigate associations among variables both within and across blocks, and (iv) to capture within-block variability in order to reveal shared and block-specific structures in a common latent space. An unified algebraic framework based on factorization strategies, denoted *canonical factorisation*, was proposed by [**?**] for multiblock methods to achieve these objectives. In practice, canonical factorisation provides a structured framework for analysing data variability at multiple levels: differences between and within blocks, differences between observations by projecting them onto a common subspace. It also facilitates the harmonisation of outputs produced by different methods and supports a more objective criterion-based selection among competing approaches.

This framework is a compass among multiblock models: Generalized Canonical Correlation Analysis (GCCA, [**?**]) and its popular case MAXVAR [**?**], Hierarchical Principal Component Analysis (HPCA, [**?**, **?**]), Structuration de Tableaux A Tois Indices de la Statistique (Statis, [**?**]), Multiple Factorial Analysis (MFA, [**?**]), Common Components and Specific Weights Analysis (CCWSA, [**?**]), Multiple CO-inertia Analysis (MCOA, [**?**]) fall within this framework. A benchmark of these methods had been done in [**?**].

Initially, MCOA was proposed as an alternative to generalized canonical analysis, also known as MAXVAR [**?**, **?**], for correlating multiple data tables. Compared with other generalisations of Hotelling's canonical correlation analysis [**?**], such as those proposed by [**?**], MCOA offers three main advantages. Firstly, it is computationally straightforward as it reduces to a singular value decomposition, the other approaches typically rely on iterative optimisation procedures. Secondly, MCOA is particularly effective at summarizing results across multiple datasets. Finally, MCOA provides sequential estimation of canonical factorisation under an orthogonality constraint. In this work, we propose a new method, Optimal Multiple CO-inertia Analysis (OMCOA), which is a

simultaneous version of MCOA. We compare this model with the previous one using simulated data whose structure is known.

2 Optimal Multiple CO-Inertia Analysis

Notations

- K blocks $(\mathbf{X}_1, \ldots, \mathbf{X}_K)$ respectively of $p_1, \ldots, p_K$ variables on N individuals.
- $\mathbf{X} = [\mathbf{X}_1 | \ldots | \mathbf{X}_K] \in \mathcal{M}_{N,P}(\mathbb{R})$ $(P = \sum_{k=1}^{K} p_k)$ be the concatenation of these K-blocks.
- $\mathbf{U}^\intercal$ designates the transposed matrix of $\mathbf{U}$.
- $\Sigma_k = diag(\sigma_k^{(1)}, ..., \sigma_k^{(R)})$ are diagonal matrices
- $\|.\|$ correspond to the Frobenius norm.

Without loss of generality, the blocks are assumed to be column-centered.

2.1 Minimization problem

[**?**] proposed a unified framework for the factorization of blocks $(\mathbf{X}_1, \ldots, \mathbf{X}_K)$ (canonical factorization). As with the singular value decomposition of a single block $\mathbf{X}$, the key insight is to determine an optimal rank-R approximation of $\mathbf{X}_1, \ldots, \mathbf{X}_K$ of the form

$$\begin{cases} \mathbf{X} = \mathbf{U}\Sigma\mathbf{V}^\intercal + \mathbf{E}, \\ \mathbf{X}_k = \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal + \mathbf{E}_k \quad \text{for} \quad k = 1, \ldots, K \end{cases} \tag{1}$$

with the constraints $\mathbf{U}^\intercal\mathbf{U} = \mathbf{I}_R$ and $\|\mathbf{v}_k^{(r)}\| = 1$ for $r = 1, \ldots, R$.

Multiblock methods provide a solution to this canonical factorization under different approximation criteria (e.g., maximization of covariance or minimization of the Frobenius norm). Projecting the data onto $\mathbf{U}$ captures the structure of the observations (for details, see [**?**]). The benchmark study in [**?**] showed that all methods under consideration yielded similar results. The decomposition of the concatenated block $\mathbf{X}$ is unique across all multiblock methods; however, only MCOA guarantees the uniqueness of the decomposition of each individual block $\mathbf{X}_k$ (by virtue of the orthogonality of the columns of $\mathbf{V}_k$, for all $k = 1, \ldots, K$). This comes at the cost of explaining a lower proportion of variance than the other methods.

2.2 Multiple CO-Inertia Analysis

[**?**] proposed Multiple Co-inertia Analysis as an alternative to MAXVAR [**?**]. The rank-H approximation is derived from successive rank-1 approximations of deflated versions of the blocks $\mathbf{X}_1, \ldots, \mathbf{X}_K$.

The rank-1 approximation criterion is based on minimising the sum of squared Frobenius norms

$$\sum_{k=1}^{K} \|\mathbf{X}_k - \sigma_k\mathbf{u}\mathbf{v}_k^\intercal\|^2, \quad \|\mathbf{u}\| = \|\mathbf{v}_1\| = \ldots = \|\mathbf{v}_K\| = 1, \quad \sigma_k > 0. \tag{2}$$

The solution to this minimisation problem is:

- $\mathbf{u}$ is the first left singular vector of the concatenated block $\mathbf{X}$,
- $\mathbf{v}_k = \frac{\mathbf{X}_k^\intercal \mathbf{u}}{\|\mathbf{X}_k^\intercal \mathbf{u}\|}$,
- $\sigma_k = \|\mathbf{X}_k^\intercal \mathbf{u}\|$ for $k = 1, \ldots, K$.

The algorithm proposed by [**?**] proceeds as follows: at each step $r = 1, \ldots, R$, the rank-1 approximation is applied to the deflated blocks $\mathbf{X}_k^{(r)} = \mathbf{X}_k(I_{p_k} - \sum_{i=1}^{r-1} \mathbf{v}_k^{(i)} \mathbf{v}_k^{(i)\intercal})$. Moreover, they proved that the resulting vectors $\mathbf{u}^{(1)}, \ldots, \mathbf{u}^{(R)}$ are mutually orthogonal. Note that, thanks to the deflation procedure, $\mathbf{v}_k^{(1)}, \ldots, \mathbf{v}_k^{(R)}$ are mutually orthogonal for each k. MCOA belongs to the class of so-called sequential multiblock component methods. In the first stage, block loadings are computed as solutions to an optimization criterion; higher-order block components are then computed from the deflated versions of the original blocks. The sequential approach is considered sub-optimal: although the parameter estimation at each stage is optimal in the least-squares sense, the corresponding criterion does not account for the full set of block components. The present study proposes an optimal alternative to MCOA.

2.3 Optimal Multiple CO-inertia Analysis (OMCOA)

Providing an optimal alternative to MCOA means minimizing

$$\sum_{k=1}^{K} \|\mathbf{X}_k - \mathbf{U}\Sigma_k \mathbf{V}_k^\intercal\|^2 = \|\mathbf{X} - \mathbf{U}\Sigma\mathbf{V}^\intercal\|^2, \tag{3}$$

with $\mathbf{V}_k^\intercal.\mathbf{V}_k = I_R$ for $k = 1, \ldots, K$. Note that, with standardized and orthogonal block-loadings matrices $\mathbf{V}_1, \ldots, \mathbf{V}_K$, the solution of this problem is unique.

Consider $\mathbf{H}_k = \Sigma_k.\mathbf{V}_k^\intercal \in \mathcal{M}_{R,p_k}(\mathbb{R})$ and $\mathbf{H} = [\mathbf{H}_1|\ldots|\mathbf{H}_K] \in \mathcal{M}_{R,P}(\mathbb{R})$. The iterative algorithm for building the solution of (**??**) requires to update successively $\mathbf{U}, \mathbf{V}_k, \Sigma_k$ as follows:

1. Updating of $\mathbf{U}$: For given $\mathbf{X} \in \mathcal{M}_{N,P}(\mathbb{R})$, $\mathbf{H} \in \mathcal{M}_{R,P}(\mathbb{R})$, update $\mathbf{U}$ by $\mathbf{U} = \mathbf{A}.\mathbf{B}^\intercal$ where $\mathbf{A}, \mathbf{B}$ are respectively the matrices of the first R left singular vectors and the matrix of the first R right singular vectors from the svd of $\mathbf{X}.\mathbf{H}^\intercal$.
2. Updating of $\mathbf{V}_k$: For given $\mathbf{X}_k \in \mathcal{M}_{N,p_k}(\mathbb{R})$, $\mathbf{U} \in \mathcal{M}_{R,N}(\mathbb{R})$ with orthogonal columns, $\Sigma_k \in \mathcal{M}_{R,R}(\mathbb{R})$, update $\mathbf{V}_k$ by: $\mathbf{V}_k = \mathbf{A}\mathbf{B}^\intercal$ where $\mathbf{A}, \mathbf{B}$ are respectively the matrix of the first R left singular vectors and the matrix of the first R right singular vectors from the svd of $\mathbf{X}_k^\intercal.\mathbf{U}$.
3. Updating of Σ_k: For given $\mathbf{X}_k \in \mathcal{M}_{N,p_k}(\mathbb{R})$, $\mathbf{U} \in \mathcal{M}_{R,N}(\mathbb{R})$ orthogonal, $\mathbf{V}_k \in \mathcal{M}_{R,p_k}(\mathbb{R})$ with orthonormal columns, Σ_k is given by

$$\Sigma_k = diag(\mathbf{U}^\intercal \mathbf{X}_k \mathbf{V}_k).$$

Proposition 1. *The successive updates of $\boldsymbol{U}$, $\boldsymbol{V}_k$, and Σ_k guarantee that the criterion defined in* (**??**) *decreases or remains constant at each step, ensuring convergence of the algorithm.*

Remark. This proof guarantees convergence of the values of the criterion, but not necessarily uniqueness of the limit point. Uniqueness of the solution is established separately, via the orthogonality of the columns of $\mathbf{V}_k$ for all k.

3 Comparison of MCOA and OMCOA

3.1 Simulation study

MCOA and OMCOA are compared on simulated data. Three blocks $\mathbf{X}_1, \mathbf{X}_2, \mathbf{X}_3$ with respectively $p_1 = 10, p_2 = 15, p_3 = 20$ variables and $I = 100$ observations have been generated from:

$$\mathbf{X}_k = \mathbf{U}\Sigma_k\mathbf{V}_k^T + \mathbf{E}_k, \quad \text{for } k = 1, \ldots, 3,$$

where $\mathbf{E}_k$ is the residual term and for $k = 1, \ldots, 3$ the columns of the matrices $\mathbf{U}_k$ (respectively $\mathbf{V}_k$) are orthonormal. We compare the estimation of $\Sigma_1, ...\Sigma_K$ obtained using MCOA and OMCOA. We consider the following simulation scenario

$$\Sigma_k(t) = (1-t)\,\Sigma_k^{(0)} + t\,\Sigma_k^{(1)}, \quad t \in [0,1].$$

where :

- For $t = 0$, all the blocks have a common profile.
  $\Sigma_1^{(0)} = \Sigma_2^{(0)} = \Sigma_3^{(0)} = \begin{pmatrix} 0.7 & & \\ & 0.5 & \\ & & 0.3 \end{pmatrix}$
- For $t = 1$ each block has its own dominant dimension of the common space $\mathbf{U}$ with permuted profiles.
  $\Sigma_1^{(1)} = \begin{pmatrix} 0.7 & & \\ & 0.5 & \\ & & 0.3 \end{pmatrix}$, $\Sigma_2^{(1)} = \begin{pmatrix} 0.5 & & \\ & 0.3 & \\ & & 0.7 \end{pmatrix}$, $\Sigma_3^{(1)} = \begin{pmatrix} 0.3 & & \\ & 0.7 & \\ & & 0.5 \end{pmatrix}$.
- $\sum_{k,r}(\sigma_k^{(r)}(0))^2 = \sum_{k,r}(\sigma_k^{(r)}(1))^2 = 0.83$.
- In all cases, $\|\mathbf{E}_k\|^2 = .10 \times \|\mathbf{X}_k\|^2$

3.2 Results

Figure **??** and **??** provide the results of the estimation by MCO and OMCOA respectively for the two scenari described in Section 5.1. As expected, the estimates in the common-configuration case were very close (see Figure **??**). In the second scenario, where all the blocks had different configurations, OMCOA outperformed MCOA in terms of both bias and precision across all parameter sets (see Figure **??**). In both scenari, the estimates obtained with OMCOA did not vary with the magnitude of the residuals.

plot1.png

plot2.png